

1.

The RL is a type of ML where an agent learns by interacting with an environment. It takes actions, receives rewards & penalties.

Main objective-

- To learn best Policy
- To maximize the total cumulative reward over time.

2.

A MDP is a mathematical model used in RL to describe decision-making where the next state depends only on the current state and action.

components of MDP

S - set of state

A - set of Action

P - state Transition Probability

R - Reward Function

3.

Bellman Equation defines the value of state as the expected reward plus the discounted value of the next state.

$$V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma V(s')]$$

for deterministic Policy

$$V(s) = R + \gamma V(s')$$

4.

exploration	exploitation
Tries new actions	choose the best known action
Gains more knowledge	Maximize immediate reward
May get lower reward initially	Gives higher short-term reward

5.

The multi-Armed Bandit Problem is a RL Problem where an agent must choose among several actions each giving different rewards.

The goal is to maximize total reward by balancing:

- exploration
- exploitation

6.

DP is a method used to solve complex Problem by breaking them into smaller subproblems

Applications in RL

- Policy Evaluation
- Policy Improvement
- Policy Iteration
- Value Iteration
- Value
- Finding Optimal Policy

7.

TD learning is a RL method that learns directly from experience by updating value after every step.

$$V(s) = V(s) + \alpha [R + \gamma V(s') - V(s)]$$

8.

SARSA	Q-Learning
on-Policy algorithm uses the action actually taken Safer Learning More conservative	off-Policy algorithm uses the best possible action Faster Learning More aggressive

9.

DQN is a RL algorithm that uses a DNN to approximate Q -value instead of Q -table

Advantages

- work with large state spaces
- Handles high-dimensional inputs
- Generalizes to unseen states
- More scalable than tabular Q -learning.

10.

DDQN is an improved version of DQN that uses two neural networks:

- online Network
- Target Network

This reduces the overestimating of Q -values.

Why is DDQN preferred over DQN

- Reduces overestimation error.
- Gives more accurate Q -value.
- Improves learning stability.
- Achieves better performance in many RL tasks.